



PHYS3071

# Introduction to Astrophysics

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Course Notes

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# Introduction

## About These Notes

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These notes accompany the PHYS3071 Astrophysics course in Hong Kong University of Science and Technology (HKUST). They are intended as a self-contained supplement to the lectures, providing both conceptual grounding and mathematical rigour at a level appropriate for readers with a background in basic physics, basic linear algebra, and basic calculus.

The material covers different astrophysical topics, including but not limited to the formation and evolution of stars, the type and spectrum of stars. Each topic is structured to serve different learning objectives—from building physical intuition to solving examination-style problems and exploring advanced derivations.

*These notes do not replace lecture slides or the recommended textbooks. They are best used alongside them as a structured study companion.*

If you find any errors or have suggestions for improvement, please feel free to contact me at [yhip@connect.ust.hk](mailto:yhip@connect.ust.hk). You can check the latest version of the notes on GitHub: <https://github.com/AuTomatoc-Ti/phys3071-Astrophysics-notes/blob/main/main.pdf>.

## Document Conventions

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### Block Components

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Each topic is presented using a consistent **5-component block structure** designed to support multiple learning styles:

#### PHYSICS INTUITION

**Purpose:** Builds conceptual understanding through plain language explanations.

This section provides the *motivation* behind the physics concept, explores its *historical development*, and offers *intuitive explanations* that help you grasp what the theory is really about before diving into equations.

*Use this when:* You're first encountering a new concept or need to refresh the "big picture" before tackling mathematics.

### THEORY

**Purpose:** Provides rigorous mathematical treatment of the concept. Here you'll find detailed *derivations*, *fundamental assumptions*, *key equations*, and connections between *mathematics and physical intuition*. This is the technical foundation required for exams and deeper problem-solving.

*Use this when:* You need to understand how to apply formulas, prepare for exams, or work through problems rigorously.

### QUESTION EXAMPLES

**Purpose:** Demonstrates practical application through worked problems.

Each question example mimics exam-style problems and includes complete *solution walkthroughs*. These help you bridge the gap between theory and problem-solving, and expose common pitfalls.

*Use this when:* Practicing problem-solving, preparing for assessments, or checking if you can apply the formulas correctly.

### FOUNDATION

**Purpose:** Supplies background knowledge and detailed proof steps that underpin the main content.

This section presents *prerequisite mathematics or physics*, *step-by-step derivations* that are too detailed to include in the Theory block, and *intermediate results* needed for a thorough understanding. Reading this is optional for the exam but essential for genuine mastery.

*Use this when:* A derivation in Theory feels too abrupt, you want to verify an intermediate step, or you need to solidify prerequisite concepts.

### EXTRA

**Purpose:** Provides supplementary material that extends *beyond* the course syllabus.

This section offers *interesting asides*, *historical context*, *connections to current research*, and citations to deeper references. The content here is intentionally out-of-scope for the course examination but enriches the broader picture of each topic.

*Use this when:* You're curious to explore beyond the syllabus, or when a topic connects to your personal research interests.

### Suggested Reading Paths:

1. **Exam Preparation:** *Physics Intuition* → *Theory* → *Question Examples*
2. **Deep Learning:** *Physics Intuition* → *Foundation* → *Theory* → *Question Examples* → *Extra*
3. **Quick Review:** *Theory* → *Question Examples*

### Special Annotations

In addition to the five block components above, two inline annotations appear throughout the text to flag specific types of information:

#### IMPORTANT NOTE

**Purpose:** Alerts you to a result, condition, or sign convention that is easy to get wrong or is frequently tested.

An Important Note signals a *common misconception*, a *critical sign*, a *non-obvious condition* (e.g. a constraint that must hold for an equation to be valid), or an *exam trap* worth memorising explicitly.

*Treat these as mandatory reading regardless of which reading path you follow.*

#### REMARK

**Purpose:** Highlights a useful observation, connection, or naming convention.

A Remark draws attention to a *conceptual link* between results, an *elegant reformulation*, a *naming convention* (e.g. why a quantity carries its particular name), or a *minor but clarifying point* that does not rise to the level of an Important Note.

*Use this when:* Looking for a deeper understanding of why an equation takes the form it does, or for useful mnemonics.

## Mathematical Conventions

### Key symbols.

| Symbol           | Meaning               | Typical units      |
|------------------|-----------------------|--------------------|
| $M_{\odot}$      | Solar mass            | kg                 |
| $R_{\odot}$      | Solar radius          | m                  |
| $L_{\odot}$      | Solar luminosity      | W                  |
| $T_{\text{eff}}$ | Effective temperature | K                  |
| $\mu$            | Mean molecular weight | —                  |
| $\rho$           | Mass density          | $\text{kg m}^{-3}$ |

#### REMARK

Unless otherwise stated, stellar quantities are quoted in solar-normalised form (for example  $M/M_{\odot}$  and  $L/L_{\odot}$ ), which makes scaling relations easier to compare across different stellar masses.

**differential notation.** We use  $\dot{x} \equiv \frac{dx}{dt}$  for time derivatives of some variables  $x(t)$ .



Astrophysics is the branch of astronomy that applies the principles of physics and chemistry to understand the nature of celestial objects and phenomena. It includes but is not limited to the study of stars, galaxies, extrasolar planets, cosmic microwave background radiation, and the interstellar medium. In this notes, we will only focus on the field of stellar astrophysics, which is the study of stars, their properties, structure, types, and evolution.

## 1.1 Stars and Stellar Properties

### PHYSICS INTUITION

In this notes, we define a star as a celestial object that is:

- (1) bounded by self-gravity under hydrostatic equilibrium, and
- (2) radiating energy generated by internal sources (e.g. nuclear fusion).

For most of the case, a star is spherical due to spherical symmetry of gravity, and mainly radiating energy by nuclear fusion in the core. However, we will also encounter some exceptions (e.g. energy generated by gravitational contraction in late stages of star evolution).

This definition shows that other celestial objects (e.g. planets, comets) are not stars as they don't fulfill all conditions. On the other hand, it allow some special conditions (e.g. brown dwarfs, white dwarfs) to be classified as stars, as they are still self-gravitating and radiating energy (though not by nuclear fusion). However, be careful some astrophysicists may not treat them as "real stars" as they don't have nuclear fusion, but we will still include them in our discussion as they are important for understanding the full picture of stellar evolution.

### REMARK

The stars are usually of spherical shape due to rotational symmetric force field of gravity, but can be oblate if they rotate rapidly.

## 1.1.1 Distance and Parallax

### PHYSICS INTUITION

Parallax is the apparent shift in position of a nearby star against the background of distant stars when observed from different positions (e.g. Earth at different points in its orbit). By measuring this shift, we can determine the distance to the star using simple trigonometry.

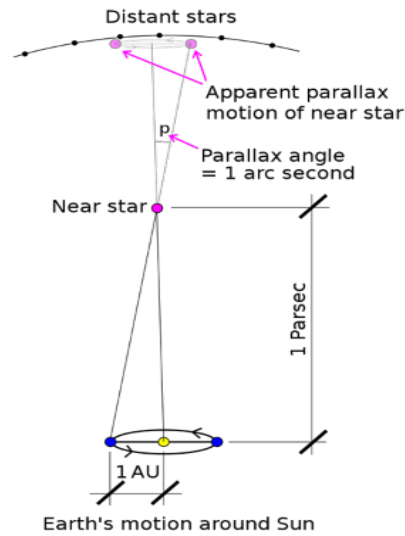


Figure 1.1: Stellar parallax illustration. The nearby star appears to shift position against the background of distant stars as Earth orbits the Sun. The angle  $p$  is the shifted angle actually, instead of the parallax angle  $\frac{1}{2}p$ .

### THEORY

Let  $x$  be the baseline (e.g. diameter of Earth's orbit),  $d$  be the distance to the star, and  $\theta$  be the parallax angle (half of the shifted angle  $p$ ). Then we have:

- Parallax distance:

$$d_{\text{pc}} = \tan^{-1} \left( \frac{x/2}{d} \right) \approx \frac{1\text{AU}}{\theta_{\text{arcsec}}} \quad (1.1)$$

- we can also directly calculate the radius of the star if we know its angular radius  $\theta$  and distance  $d$ :

$$R = d \tan \theta \approx d \theta \quad (1.2)$$

### IMPORTANT NOTE

Don't mixed up the parallax angle  $\theta$  and the shifted angle  $p$  in the parallax measurement. The total shifted angle  $p$  shows the apparent shift of the star, but when we use the trigonometry to calculate the distance, we need to use the parallax angle  $\theta = \frac{1}{2}p$  which is the angle between the line of sight to the star

and the line connecting the Earth and the Sun.

### Question Example 1.1

From the previous example, if the star has an angular diameter of 0.01 arcseconds, estimate its radius in units of  $R_{\odot}$ .

**Solution.**

$$\begin{aligned} R &= \frac{1}{2}\theta d = \frac{1}{2} \times 0.01'' \times 100 \text{ pc} \\ &= 0.5 \times 0.01 \times 100 \times 206265 \text{ AU} \\ &= 1031.325 \text{ AU} \\ &= 1031.325 \times 215 R_{\odot} \\ &\approx 221,000 R_{\odot}. \end{aligned}$$

### Question Example 1.2

Assume a telescope angular resolution of 0.01 arcseconds.

(a). what is the maximum distance at which we can resolve a star with a radius of  $R_{\odot}$ ?

(b). If baseline  $x$  is 2 AU, what is the maximum distance of a star we can resolve?

**Solution.** (a). We can use the formula for the radius of the star:

$$\begin{aligned} R &\geq d\theta \\ d &\leq \frac{R}{\theta} = \frac{R_{\odot}}{0.01''} \end{aligned}$$

As  $R_{\odot} \approx 6.957 \times 10^8 \text{ m}$  and  $1'' = 4.848 \times 10^{-6} \text{ radians}$ , we have:

$$\begin{aligned} d &\leq \frac{6.957 \times 10^8 \text{ m}}{0.01 \times 4.848 \times 10^{-6}} \\ &\approx 1.43 \times 10^{16} \text{ m} \\ &\approx 0.465 \text{ pc}. \end{aligned}$$

(b). Using the parallax distance formula:

$$\begin{aligned} d &\leq \frac{x/2}{\tan \theta} \approx \frac{1 \text{ AU}}{0.01''} \\ &\approx 100 \text{ pc}. \end{aligned}$$

### REMARK

As we can see from the above examples, the stellar parallax method is only effective for measuring distances to nearby stars (within a few hundred parsecs). For more distant stars, we need to use other methods such as spectroscopic

parallax or standard candles.(we will discuss these methods in later chapters).

### 1.1.2 Brightness, Magnitude, and Luminosity

#### THEORY

$$M_1 - M_2 = -2.5 \log_{10} \left( \frac{L_1}{L_2} \right), \quad (1.3)$$

$$m_1 - m_2 = -2.5 \log_{10} \left( \frac{F_1}{F_2} \right), \quad (1.4)$$

$$m - M = 5 \log_{10}(d/\text{pc}) - 5, \quad (1.5)$$

$$F = \frac{L}{4\pi d^2}. \quad (1.6)$$

#### Question Example 1.3: Distance modulus quick use

A star has  $m = 9.0$  and  $M = 4.0$ . Estimate its distance.

**Solution.**

$$\begin{aligned} m - M &= 5 \log_{10}(d/\text{pc}) - 5 = 5 \\ \Rightarrow \log_{10}(d/\text{pc}) &= 2 \\ \Rightarrow d &= 10^2 \text{ pc} = 100 \text{ pc}. \end{aligned}$$

### 1.1.3 Mass

## 1.2 Thermal Radiation and Blackbody Spectrum

### 1.2.1 Planck function, Rayleigh-Jeans law, Wien's law

#### THEORY

- Planck function:

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1} \quad (1.7)$$

- Stefan–Boltzmann law:

$$F = \sigma T^4, \quad \sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4} \quad (1.8)$$

- Luminosity:

$$L = 4\pi R^2 \sigma T_{\text{eff}}^4 \quad (1.9)$$

- Wien's law:

$$\lambda_{\text{max}} T = 2.9 \times 10^{-3} \text{ m K} \quad (1.10)$$

#### Question Example 1.4

A star has a peak wavelength of 500 nm. Estimate its surface temperature.

**Solution.** From Wien's law,

$$\begin{aligned} \lambda_{\text{max}} T &= 2.9 \times 10^{-3} \text{ m K} \\ T &= \frac{2.9 \times 10^{-3} \text{ m K}}{\lambda_{\text{max}}} \\ &= \frac{2.9 \times 10^{-3} \text{ m K}}{500 \times 10^{-9} \text{ m}} \\ &= 5800 \text{ K.} \end{aligned}$$

### 1.2.2 Intensity, Flux, and Luminosity

#### PHYSICS INTUITION

Observed starlight is measured in layers: direction-resolved intensity, surface flux, and finally total emitted power. Keeping these definitions separate avoids unit mistakes in later derivations.

#### THEORY

- Specific intensity  $I_\nu$ :

$$dE = I_\nu \cos \theta dA dt d\nu d\Omega \quad (1.11)$$

- Monochromatic flux density:

$$F_\nu = \int I_\nu \cos \theta \, d\Omega \quad (1.12)$$

- Bolometric flux:

$$F = \int_0^\infty F_\nu \, d\nu \quad (1.13)$$

- Luminosity:

$$L = \int F \, dA \quad (1.14)$$

#### IMPORTANT NOTE

$I_\nu$  is invariant along a ray in vacuum (no absorption/emission), while  $F_\nu$  depends on source geometry and distance. Do not interchange them in magnitude or distance calculations.

### 1.2.3 Radiative Transfer

#### Question Example 1.5: Calculating Brightness

Suppose a star is at 20 pc. How bright does it appear if its luminosity is  $100L_\odot$ ?  
**Solution.**

$$\begin{aligned} F &= \frac{L}{4\pi d^2} = \frac{100L_\odot}{4\pi(20 \text{ pc})^2} \\ &= \frac{100 \times 3.828 \times 10^{26} \text{ W}}{4\pi(20 \times 3.086 \times 10^{16} \text{ m})^2} \\ &\approx 3.1 \times 10^{-9} \text{ W/m}^2. \end{aligned}$$

### 1.2.4 Surface Temperature

## 1.3 Surface Temperature and Spectral Classification

Recall that the surface temperature  $T_{\text{eff}}$  of a star can be estimated from its luminosity and radius using the Stefan-Boltzmann law:

$$L = 4\pi R^2 \sigma T_{\text{eff}}^4 \quad (1.15)$$

where  $\sigma$  is the Stefan-Boltzmann constant. Rearranging this gives:

$$T_{\text{eff}} = \left( \frac{L}{4\pi R^2 \sigma} \right)^{1/4} \quad (1.16)$$

#### Question Example 1.6: Estimating Surface Temperature

A star has a luminosity of  $300L_{\odot}$  and a radius of  $25R_{\odot}$ . Estimate its surface temperature.

**Solution.** First, we need to convert the luminosity and radius into SI units:

$$L = 300L_{\odot} = 300 \times 3.828 \times 10^{26} \text{ W} = 1.1484 \times 10^{29} \text{ W},$$

$$R = 25R_{\odot} = 25 \times 6.957 \times 10^8 \text{ m} = 1.73925 \times 10^{10} \text{ m}.$$

Now we can plug these values into the formula for  $T_{\text{eff}}$ :

$$T_{\text{eff}} = \left( \frac{1.1484 \times 10^{29} \text{ W}}{4\pi (1.73925 \times 10^{10} \text{ m})^2 \times 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4} \right)^{1/4} \\ \approx 4500 \text{ K}.$$

### 1.3.1 Spectral Classification

Apart from direct use of  $L$ , we can also estimate  $T_{\text{eff}}$  from the star's spectral type, which is determined by its surface temperature and spectral features.

#### THEORY

Stars are classified into spectral types O, B, A, F, G, K, M based on their surface temperatures and spectral features. O-type stars are the hottest and most massive, while M-type stars are the coolest and least massive. Each spectral type corresponds to a specific range of surface temperatures, which can be used to estimate  $T_{\text{eff}}$  if the spectral type is known.

## 2.1 Sources of Star Formation

### 2.1.1 Molecular Clouds

#### PHYSICS INTUITION

Molecular clouds are dense regions of gas and dust in the interstellar medium where star formation occurs. They are primarily composed of molecular hydrogen ( $\text{H}_2$ ) and can have masses ranging from a few solar masses to millions of solar masses. The high density and low temperature of molecular clouds allow them to collapse under their own gravity, leading to the formation of stars.

## 2.2 Analysis of Formation Criteria

### 2.2.1 Hydrostatic Equilibrium

#### FOUNDATION

Assume spherical symmetry and radial dependence only. Neglect rotation, magnetic fields, and turbulence in the first model to isolate gravity-versus-pressure physics.

#### THEORY

$$dm(r, t) = 4\pi r^2 \rho dr, \quad (2.1)$$

$$\frac{\partial m}{\partial t} + v \frac{\partial m}{\partial r} = 0. \quad (2.2)$$



**Question Example 2.1**

Derive the radial acceleration  $\ddot{r}$  of a fluid element at radius  $r$  in a star, given the mass enclosed within  $r$  is  $m(r)$  and the density is  $\rho(r)$ .

**Solution.** The gravitational force per unit mass at radius  $r$  is given by:

$$g = \frac{Gm(r)}{r^2}. \quad (2.3)$$

The pressure gradient force per unit mass is given by:

$$f_P = -\frac{1}{\rho} \frac{dP}{dr}. \quad (2.4)$$

The total radial acceleration  $\ddot{r}$  is the sum of these two forces:

$$\ddot{r} = f_P + g = -\frac{1}{\rho} \frac{dP}{dr} + \frac{Gm(r)}{r^2}. \quad (2.5)$$

**Question Example 2.2**

Consider a star of mass  $M$  and radius  $R$  in hydrostatic equilibrium. If the star is suddenly compressed to half its radius, what is the radial acceleration  $\ddot{r}$  at the surface?

**Solution.** The gravitational force per unit mass at the surface is given by:

$$g = \frac{GM}{R^2}. \quad (2.6)$$

If the star is compressed to half its radius, the new radius is  $R' = R/2$ . The new gravitational force per unit mass at the surface is:

$$g' = \frac{GM}{(R/2)^2} = \frac{GM}{R^2/4} = 4 \frac{GM}{R^2} = 4g. \quad (2.7)$$

The radial acceleration  $\ddot{r}$  at the surface is then:

$$\ddot{r} = g' - g = 3g = 3 \frac{GM}{R^2}. \quad (2.8)$$

**THEORY**

**Hydrostatic equilibrium** At hydrostatic equilibrium, the inward gravitational force is balanced by the outward pressure gradient and  $\ddot{r} = 0$ :

$$\frac{dP}{dr} = -\frac{Gm(r)\rho}{r^2} \quad (2.9)$$

**Question Example 2.3**

Consider a sphere of mass  $M$  and radius  $R$  with a uniform density  $\rho$ .

- (a) Write the pressure gradient  $-\frac{dP}{dr}$  at distance  $r$  from the center at hydrostatic equilibrium.
- (b) Calculate the pressure  $P(r)$  at distance  $r$  from the center.

**Solution.**

- (a) The mass enclosed within radius  $r$  is given by:

$$m(r) = \frac{4}{3}\pi r^3 \rho. \quad (2.10)$$

The hydrostatic equilibrium equation is:

$$-\frac{dP}{dr} = \frac{Gm(r)\rho}{r^2}. \quad (2.11)$$

Substituting  $m(r)$ , we get:

$$-\frac{dP}{dr} = \frac{G\left(\frac{4}{3}\pi r^3 \rho\right)\rho}{r^2} = \frac{4\pi G}{3}r\rho^2. \quad (2.12)$$

- (b) To find  $P(r)$ , we integrate the pressure gradient from the surface ( $r = R$ , where  $P(R) = 0$ ) to a point inside the sphere:

$$\begin{aligned} P(r) &= - \int_r^R -\frac{dP}{dr'} dr' = \int_r^R \frac{4\pi G}{3} r' \rho^2 dr' \\ &= \frac{4\pi G}{3} \rho^2 \int_r^R r' dr' = \frac{4\pi G}{3} \rho^2 \left[ \frac{r'^2}{2} \right]_r^R \\ &= \frac{4\pi G}{3} \rho^2 \left( \frac{R^2 - r^2}{2} \right) = 2\pi G \rho^2 (R^2 - r^2). \end{aligned}$$

## 2.3 Virial Theorem and Jeans Criterion

### 2.3.1 Energetics of Self-Gravitating Gas

## THEORY

$$E_{\text{GR}} = - \int_0^M \frac{Gm}{r} dm = -f_{\text{GR}} \frac{GM^2}{R}, \quad (2.13)$$

$$\langle P \rangle = -\frac{1}{3} \frac{E_{\text{GR}}}{V}, \quad (2.14)$$

$$P = \frac{1}{3} nm \langle v^2 \rangle = \frac{2}{3} n \langle E_k \rangle. \quad (2.15)$$

For a star with a uniform density profile (i.e.,  $\rho = \text{constant}$ ),  $f_{\text{GR}} \approx 3/5$ .

## 2.3.2 Jeans Criterion

## PHYSICS INTUITION

The Jeans criterion determines whether a cloud of gas will collapse under its own gravity to form a star. It compares the gravitational energy of the cloud to its thermal energy. If the gravitational energy exceeds twice the thermal energy, the cloud is unstable and will collapse.

## 2.3.3 Jeans Instability

## THEORY

$$|E_{\text{GR}}| > 2E_{\text{th}} \quad (\text{collapse}), \quad (2.16)$$

$$M_J = \left( \frac{5k_B T}{G\mu m_H} \right)^{3/2} \left( \frac{3}{4\pi\rho} \right)^{1/2}, \quad (2.17)$$

$$\lambda_J = \sqrt{\frac{\pi c_s^2}{G\rho}}, \quad (2.18)$$

$$M_J \propto T^{3/2} \rho^{-1/2}. \quad (2.19)$$

**IMPORTANT NOTE**

Lower temperature and higher density both reduce  $M_J$ . This is why cooling strongly promotes fragmentation and cluster formation.

## 2.4 End of Formation: Protostars and Pre-Main Sequence Stars

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### 3.1 Statistical Distributions

#### 3.1.1 Energy-Momentum Relation

To calculate the value of various thermodynamic quantities, we need first clarify some basic concepts and notations first for convenience.

In below contents, we will use the  $E(p)$  notation to represent the energy of a particle with momentum  $p$  and mass  $m$ .

##### FOUNDATION

Recall special relativity, the energy of a particle with mass  $m$  and three-momentum  $p$  is given by

$$E(\vec{p}) = \sqrt{|\vec{p}|^2 c^2 + m^2 c^4}.$$

Taken  $c = 1$  and  $p = |\vec{p}|$ , we have  $E(p) = \sqrt{p^2 + m^2}$ .

### 3.2 Statistical Distributions

##### THEORY

$$\text{Maxwell-Boltzmann : } f_{\text{MB}}(E) = \frac{2\pi}{(k_B T)^{3/2}} \sqrt{E} e^{-E/k_B T}, \quad (3.1)$$

$$\text{Fermi-Dirac : } f_{\text{FD}}(E) = \frac{1}{e^{(E-\mu)/k_B T} + 1}, \quad (3.2)$$

$$\text{Bose-Einstein : } f_{\text{BE}}(E) = \frac{1}{e^{(E-\mu)/k_B T} - 1}. \quad (3.3)$$

In a Non-relativistic limit ( $p \ll m$ ):

$$E(p) = m \sqrt{1 + \frac{p^2}{m^2}} \approx m + \frac{p^2}{2m}.$$

In an Ultra-relativistic limit ( $p \gg m$ ):

$$E(p) \approx p.$$

### 3.2.1 Chemical Potential

#### PHYSICS INTUITION

Imagine we have a box of gas, and we want to add a particle into the box. The chemical potential  $\mu$  is the energy cost of adding that particle to the system, which can be thought of as the "price" we need to pay to add one more particle to the system.

However, when we add a particle to the system, we also increase the entropy of the system, which can be thought of as the "benefit" we get from adding that particle, since it allows for more possible microstates and thus increases the disorder of the system.

Furthermore, the temperature  $T$  of the system determines how much we value that increase in entropy, since at higher temperatures, we need to pay more energy to achieve the same increase in entropy.

Therefore, the chemical potential can be thought of as a balance between the energy cost of adding a particle and the entropy gain from adding that particle, which is given by the formula:

$$\mu = \text{energy cost} - T \cdot \text{entropy gain}$$

#### FOUNDATION

$\mu$  is the chemical potential, which measures the change in internal energy when a particle is added to the system at constant entropy and volume. It is defined as:

$$\mu = \left( \frac{\partial U}{\partial N} \right)_{S,V} = \left( \frac{\partial F}{\partial N} \right)_{T,V},$$

Or in other way:

$$\delta\mu = \delta U - T\delta S,$$

where  $U$  is the internal energy,  $F$  is the Helmholtz free energy,  $N$  is the number of particles,  $S$  is the entropy, and  $T$  is the temperature.

For ideal gases,  $\mu \rightarrow 0$ , and we will assume  $\mu = 0$  in below contents unless otherwise specified.

**REMARK**

The chemical potential  $\mu$  is typically zero for photons and other massless bosons, while it can be non-zero for massive particles depending on the context (e.g., in a gas of electrons).

while for particles and their own anti-particles,  $\mu_+ = -\mu_-$ .

### 3.2.2 Thermal Equilibrium Distributions

#### FOUNDATION

For a state with energy  $E$ , and particle number  $N$ , the probability of the state is given by the Boltzmann factor:

$$P \propto e^{-(E-\mu N)/k_B T}$$

While for a single-particle state with energy  $E(p)$ , each momentum state can have occupation number  $n$ :

- Fermions:  $n = 0$  or  $1$ ,
- Bosons:  $n = 0, 1, 2, \dots$

Then the average occupation number for a single-particle state with energy  $E(p)$  is given by:

$$\langle n \rangle = \frac{\sum_n n e^{-(E(p)-\mu n)/k_B T}}{\sum_n e^{-(E(p)-\mu n)/k_B T}}$$

For fermions, we have:

$$\langle n \rangle = \frac{0 \cdot e^{-E(p)/k_B T} + 1 \cdot e^{-(E(p)-\mu)/k_B T}}{e^{-E(p)/k_B T} + e^{-(E(p)-\mu)/k_B T}} = \frac{1}{e^{(E(p)-\mu)/k_B T} + 1}$$

For bosons, we have:

$$\langle n \rangle = \frac{\sum_{n=0}^{\infty} n e^{-(E(p)-\mu n)/k_B T}}{\sum_{n=0}^{\infty} e^{-(E(p)-\mu n)/k_B T}} = \frac{1}{e^{(E(p)-\mu)/k_B T} - 1}$$

By letting  $f(p) = \langle n \rangle$ , we can calculate the number density by density of states  $\times$  occupation number, which is given by:

$$\frac{dN}{d^3p} = \overbrace{\left( \frac{g}{(2\pi)^3} \right)}^{\text{density of states}} \overbrace{f(p)}^{\text{occupation number}} = \frac{g}{(2\pi)^3} \frac{1}{e^{(E(p)-\mu)/k_B T} \pm 1}$$

The term  $\frac{g}{2\pi^2}$  come from the Heisenberg uncertainty principle, which states that density states, i.e. the number of quantum states in a given volume of phase space as  $(2\pi\hbar)^3$  under natural unit, and the factor  $g$  accounts for the spin degeneracy of the particle species.



**THEORY**

In thermal equilibrium, the distribution of particle energies follows specific statistical distributions:

- **Fermi-Dirac distribution** for fermions:

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_B T} + 1}$$

- **Bose-Einstein distribution** for bosons:

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_B T} - 1}$$

- **Maxwell-Boltzmann distribution** for classical particles:

$$f(p) = e^{-(E(p)-\mu)/k_B T}$$

In general, the distribution can be written as:

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_B T} \pm 1},$$

where the + sign is for fermions and the – sign is for bosons.

**THEORY**

In non-relativistic limit ( $E \gg T$ ), the Fermi-Dirac and Bose-Einstein distributions reduce to the Maxwell-Boltzmann distribution:

$$e^{(E(p)-\mu)/k_B T} \gg 1 \implies f(p) \approx e^{-(E(p)-\mu)/k_B T}$$

### 3.3 Number Density, Energy Density, and Pressure

#### 3.3.1 Spin Degeneracy Factor

##### PHYSICS INTUITION

Imagine in a hotel, there are multiple rooms with different price (Energy level), the spin is like the no. of beds in the room, which can accommodate multiple guests (particles) with the same energy level. When 2 particles enter the same room (share the same energy level), we call it a degenerate state.

The degeneracy factor  $g$  states how many particles can stay in the same energy level, which is determined by the intrinsic spin quantum number  $s$  of the particle. On the other hand, it can be treated as the degree of freedom of the particle, which is the number of independent ways the particle can exist in a given energy state.

##### FOUNDATION

In quantum mechanics, the magnetic quantum number  $m_s$  can take on values from  $-s$  to  $+s$  in integer steps ( $-s, -s + 1, \dots, s - 1, s$ ). And we can get  $g$  by counting the number of possible  $m_s$  values, which is given by:

$$g = 2s + 1$$

However, for massless particles, such as photons, the spin is always aligned with the direction of motion, which means they only have 2 possible spin states (helicity states), thus  $g = 2$  for photons.

##### REMARK

The  $g$  here only counts the intrinsic spin degeneracy, but in general, we also need to consider other types of degeneracy, such as color charge for quarks, which can further increase the effective degeneracy factor for those particles: For example, if we treat the 8 type of gluons as the same one and calculate the  $g$  for gluons, we will get  $g_{\text{total}} = 16$ .

$$g_{\text{effective}} = g_{\text{spin}} \times g_{\text{color}}$$

In the below contents, when we see  $g_i$ , it only represents the intrinsic spin degeneracy, and we will need to multiply it with other degeneracy factors if there are any.

**EXTRA**

The  $g$  values for some common particles are:

| Particle          | symbol         | $g$ |
|-------------------|----------------|-----|
| Photon            | $g_\gamma$     | 2   |
| Electron/Positron | $g_e^-, g_e^+$ | 2   |
| Z boson           | $g_Z$          | 3   |
| Higgs boson       | $g_H$          | 1   |

### 3.3.2 Derivation of the 3 equations

**THEORY**

The number density  $n$ , energy density  $\rho$ , and pressure  $P$  of some particle  $i$  are given by:

$$n_i = \frac{g_i}{(2\pi)^3} \int f_i(p) d^3p,$$

$$\rho_i = \frac{g_i}{(2\pi)^3} \int E(p) f_i(p) d^3p,$$

$$P_i = \frac{g_i}{(2\pi)^3} \int \frac{p^2}{3E(p)} f_i(p) d^3p$$

By substituting the Fermi-Dirac, Bose-Einstein, or Maxwell-Boltzmann distribution functions into the above integrals, we can derive explicit expressions for  $n_i$ ,  $\rho_i$ , and  $P_i$  for different particle species under various conditions (e.g., relativistic vs. non-relativistic limits).

**FOUNDATION****(1) Relativistic Fermions ( $E \approx p$ ):**

- Number density:

$$\begin{aligned}
 n_i &= \frac{g_i}{(2\pi)^3} \int \frac{d^3p}{e^{(E(p)-\mu)/k_B T} + 1} \\
 &= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{4\pi p^2 dp}{e^{(p-\mu)/k_B T} + 1} \quad , \text{ using spherical coordinates} \\
 &= \frac{g_i}{2(\pi)^2} k_B^3 T^3 \int_0^\infty \frac{u^2 du}{e^u + 1} \quad , u = \frac{p - \mu}{k_B T} \\
 &= \frac{g_i}{2(\pi)^2} k_B^3 T^3 \cdot \frac{3}{2} \zeta(3) \\
 &= \frac{3}{4} \frac{\zeta(3)}{\pi^2} g_i T^3
 \end{aligned}$$

- Energy density:

$$\begin{aligned}
 \rho_i &= \frac{g_i}{(2\pi)^3} \int E(p) f_i(p) d^3p \\
 &= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{p \cdot 4\pi p^2 dp}{e^{(p-\mu)/k_B T} + 1} \quad , E = p \text{ for relativistic particles} \\
 &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \int_0^\infty \frac{u^3 du}{e^u + 1} \quad , u = \frac{p - \mu}{k_B T} \\
 &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \cdot \frac{7}{8} \cdot 6\zeta(4) \\
 &= \frac{7}{8} \frac{\pi^2}{30} g_i T^4
 \end{aligned}$$

- Pressure:  
using:

$$\frac{p^2}{3E(p)} = \frac{1}{3} \frac{p^2}{p} = \frac{1}{3} p,$$

we get:

$$\begin{aligned}
 P_i &= \frac{g_i}{(2\pi)^3} \int \frac{p^2}{3E(p)} f_i(p) d^3p \\
 &= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{\frac{1}{3} p \cdot 4\pi p^2 dp}{e^{(p-\mu)/k_B T} + 1} \\
 &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \int_0^\infty \frac{u^3 du}{e^u + 1} \quad , u = \frac{p - \mu}{k_B T} \\
 &= \frac{7}{8} \frac{\pi^2}{30} g_i T^4 \cdot \frac{1}{3} \\
 &= \frac{1}{3} \rho_i
 \end{aligned}$$

**(2) Relativistic Bosons ( $E \approx p$ ):**

- Number density:

$$\begin{aligned}
 n_i &= \frac{g_i}{(2\pi)^3} \int \frac{d^3p}{e^{(E(p)-\mu)/k_B T} - 1} \\
 &= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{4\pi p^2 dp}{e^{(p-\mu)/k_B T} - 1} \\
 &= \frac{g_i}{2(\pi)^2} k_B^3 T^3 \int_0^\infty \frac{u^2 du}{e^u - 1} \quad , u = \frac{p - \mu}{k_B T} \\
 &= \frac{g_i}{2(\pi)^2} k_B^3 T^3 \cdot 2\zeta(3) \\
 &= \frac{\zeta(3)}{\pi^2} g_i T^3
 \end{aligned}$$

- Energy density:  
using  $E(p) \approx p$ , we have:

$$\begin{aligned}
 \rho_i &= \frac{g_i}{(2\pi)^3} \int E(p) f_i(p) d^3p \\
 &= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{p \cdot 4\pi p^2 dp}{e^{(p-\mu)/k_B T} - 1} \\
 &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \int_0^\infty \frac{u^3 du}{e^u - 1} \quad , u = \frac{p - \mu}{k_B T} \\
 &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \cdot 6\zeta(4) \\
 &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \cdot \frac{\pi^4}{15} \\
 &= \frac{\pi^2}{30} g_i T^4
 \end{aligned}$$

- Pressure:

$$\begin{aligned}
 P_i &= \frac{g_i}{(2\pi)^3} \int \frac{p^2}{3E(p)} f_i(p) d^3p \\
 &= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{\frac{1}{3}p \cdot 4\pi p^2 dp}{e^{(p-\mu)/k_B T} - 1} \\
 &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \int_0^\infty \frac{u^3 du}{e^u - 1} \quad , u = \frac{p - \mu}{k_B T} \\
 &= \frac{\pi^2}{30} g_i T^4 \cdot \frac{1}{3} \\
 &= \frac{1}{3} \rho_i
 \end{aligned}$$

**(3) Non-relativistic Particles ( $E \approx m + p^2/2m$ ):**

- Number density:

$$\begin{aligned}
 n_i &= \frac{g_i}{(2\pi)^3} \int e^{-(E(p)-\mu)/k_B T} d^3p \\
 &= \frac{g_i}{(2\pi)^3} e^{-(m-\mu)/k_B T} \int e^{-p^2/2mk_B T} d^3p \\
 &= \frac{g_i}{(2\pi)^3} e^{-(m-\mu)/k_B T} \int 4\pi p^2 e^{-p^2/2mk_B T} dp \\
 &= \frac{g_i}{2(\pi)^2} e^{-(m-\mu)/k_B T} \cdot \frac{\sqrt{\pi}}{4} (2mk_B T)^{3/2} \quad , \text{ using Gaussian integral} \\
 &= g_i \left( \frac{mk_B T}{2\pi} \right)^{3/2} e^{-(m-\mu)/k_B T}
 \end{aligned}$$

- Energy density: using  $E(p) \approx m + \frac{p^2}{2m}$ , we have:

$$\begin{aligned}
 \rho_i &= \frac{g_i}{(2\pi)^3} \int E(p) f_i(p) d^3p \\
 &\approx \frac{g_i}{(2\pi)^3} \int \left( m + \frac{p^2}{2m} \right) e^{-(E(p)-\mu)/k_B T} d^3p \\
 &= \frac{g_i m}{(2\pi)^3} e^{-(m-\mu)/k_B T} \int e^{-p^2/2mk_B T} d^3p \\
 &\quad + \frac{g_i}{(2\pi)^3} e^{-(m-\mu)/k_B T} \int \frac{p^2}{2m} e^{-p^2/2mk_B T} d^3p \\
 &= n_i m + \frac{3}{2} n_i k_B T \quad , \text{ using Gaussian integral} \\
 &= n_i m \quad , \text{ since } m \gg T \text{ for non-relativistic particles}
 \end{aligned}$$

- Pressure: using  $\frac{p^2}{3E(p)} \approx \frac{p^2}{3m}$ , we have:

$$\begin{aligned}
 P_i &= \frac{g_i}{(2\pi)^3} \int \frac{p^2}{3E(p)} f_i(p) d^3p \\
 &\approx \frac{g_i}{(2\pi)^3} \cdot \frac{2}{3} \int \frac{p^2}{2m} e^{-(E(p)-\mu)/k_B T} d^3p \\
 &= n_i T \quad , \text{ by using the result from energy density calculation}
 \end{aligned}$$

**THEORY**

Number density, energy density, and pressure for particles with different thermal equilibrium distributions:

|                         | Relativistic Fermions                        | Relativistic Bosons              | Non-relativistic Particles (Fermions/Bosons)             |
|-------------------------|----------------------------------------------|----------------------------------|----------------------------------------------------------|
| Number Density $n_i$    | $\frac{3}{4} \frac{\zeta(3)}{\pi^2} g_i T^3$ | $\frac{\zeta(3)}{\pi^2} g_i T^3$ | $g_i \left( \frac{m_i T}{2\pi} \right)^{3/2} e^{-m_i/T}$ |
| Energy Density $\rho_i$ | $\frac{7}{8} \frac{\pi^2}{30} g_i T^4$       | $\frac{\pi^2}{30} g_i T^4$       | $n_i m_i$                                                |
| Pressure $P_i$          | $\frac{1}{3} \rho_i$                         | $\frac{1}{3} \rho_i$             | $n_i T$                                                  |

where  $\zeta(3) \approx 1.202$  is the Riemann zeta function evaluated at 3.

We can see once the particles no longer relativistic, the number density and energy density will be exponentially suppressed.

It is because in the thermal plasma, the energy of particle are  $E \sim T$ , and when  $T \ll 2m$ , the probability of pair production which require energy  $E \geq 2m$  will be very low, thus the number density and energy density will be suppressed.

**PHYSICS INTUITION**

It also explains why very few positrons exist in the current universe! In the early universe,  $T \gg m_e \approx 0.511 \text{ MeV}$ ,  $e^+e^-$  pairs were abundant. As the universe cooled below  $T \sim m_e$ , the number density of positrons (and electrons) dropped exponentially, leaving only a tiny fraction that survived annihilation, which is why we observe so few positrons today.

**PHYSICS INTUITION**

We can also observed the fermions have lower number density and energy density than bosons.

Actually it also match our expectation in previous section, we know that bosons can occupied the same quantum state but fermions cannot (Pauli exclusion principle), thus we can expect that fermions have lower number density and energy density than bosons as lower occupation number.

**PHYSICS INTUITION**

For non-relativistic particles, we can see the energy density is dominated by the rest mass energy  $n_i m_i$ , while the pressure is given by  $n_i T$ , which is just the ideal gas law  $P = nk_B T$  with  $k_B = 1$  under natural unit.

**THEORY**

We can see the relationship between  $p$  and  $\rho$  for different relativistic limits:

$$P = \begin{cases} \frac{1}{3}\rho & , T \gg m \\ nT \ll \rho & , T \ll m \end{cases}$$

In fact, it also consistent with the assumption of state parameter of radiation and matter, which are  $w = 1/3$  and  $w = 0$  when in relativistic and non-relativistic limits respectively.

**THEORY**

The average particle energy is given by:

$$\langle E \rangle = \frac{\rho_i}{n_i} = \begin{cases} \frac{7\pi^4}{180\zeta(3)} T \approx 3.151T & , T \gg m \text{ for relativistic fermions} \\ \frac{\pi^4}{30\zeta(3)} T \approx 2.701T & , T \gg m \text{ for relativistic bosons} \\ m + \frac{3}{2}T & , T \ll m \text{ for non-relativistic particles} \end{cases}$$

Thats mean we can have a simplification of  $\langle E \rangle \sim T$  for relativistic particles, as they have same order of magnitude, which is low enough error in astronomy :)

### 3.4 Equation of State (EOS) of Stellar Matter

**PHYSICS INTUITION**

The equation of state (EOS) relates the pressure, density, and temperature of a substance. For stellar matter, the EOS is crucial for understanding the internal structure and evolution of stars.

**THEORY**

$$P = P(\rho, T, \{X_i\}), \quad (3.4)$$

$$\text{Ideal gas : } P = \frac{\rho k_B T}{\mu m_H}, \quad (3.5)$$

$$\text{Radiation pressure : } P_{\text{rad}} = \frac{1}{3} a T^4, \quad a = \frac{4\sigma}{c}, \quad (3.6)$$

$$\text{Degenerate gas : } P_{\text{deg}} \propto \rho^{5/3} \quad (\text{non-relativistic}), \quad (3.7)$$

$$P_{\text{deg}} \propto \rho^{4/3} \quad (\text{relativistic}). \quad (3.8)$$



## 3.5 Radiation

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### THEORY

$$u_{\text{rad}} = aT^4, \quad (3.9)$$

$$F_{\text{rad}} = \frac{c}{4}u_{\text{rad}} = \sigma T^4, \quad (3.10)$$

$$L_{\text{rad}} = 4\pi R^2 F_{\text{rad}} = 4\pi R^2 \sigma T^4. \quad (3.11)$$

Three main mechanisms of heat transfer in stars:

1. Conduction
2. Convection
3. Radiation (by photons and neutrinos from core)

## 4.1 Conduction

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## 4.2 Convection

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### 4.2.1 Convection Mechanism

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## 4.3 Radiation

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### 4.3.1 Scattering

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### 4.3.2 Mean Free Path

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#### THEORY

The mean free path  $\lambda$  of a photon in a medium is given by:

$$\lambda = \frac{1}{n\sigma}, \quad (4.1)$$

where  $n$  is the number density of scatterers and  $\sigma$  is the scattering cross-section.

Recall that there are three primary energy sources in stars: gravitational contraction, radioactive decay and nuclear fusion.

In this chapter, we will focus on the process of thermonuclear fusion, which is the viable long-term stellar energy source in most stars.

## 5.1 Requisites for Thermonuclear Fusion

### 5.1.1 Nuclear Size and Binding Energy

#### THEORY

$$R = R_0 A^{1/3}, \quad R_0 \approx 1.2 \times 10^{-15} \text{ m}, \quad (5.1)$$

$$BE(Z, N) = [Zm_p + Nm_n - M(Z, N)] c^2. \quad (5.2)$$

Binding energy per nucleon increases for light nuclei and decreases for very heavy nuclei due to Coulomb repulsion.

#### Question Example 5.1

Calculate the binding energy per nucleon for helium-4 ( ${}^4\text{He}$ ) and  ${}^{12}\text{C}$ .

**Solution.**

$$BE({}^4\text{He}) = [2m_p + 2m_n - M({}^4\text{He})] c^2 \approx 28.3 \text{ MeV},$$

$$BE({}^4\text{He})/4 \approx 7.1 \text{ MeV/nucleon},$$

$$BE({}^{12}\text{C}) = [6m_p + 6m_n - M({}^{12}\text{C})] c^2 \approx 92.2 \text{ MeV},$$

$$BE({}^{12}\text{C})/12 \approx 7.7 \text{ MeV/nucleon}.$$

### 5.1.2 Why Fusion Powers Stars

#### PHYSICS INTUITION

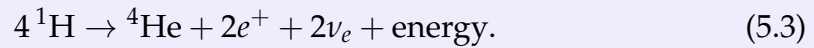
Stars gain energy when fusing light nuclei toward higher binding energy per nucleon. This trend reverses near iron, where further fusion is endothermic.

## 5.2 Hydrogen and Helium Burning

### 5.2.1 Proton-Proton (p-p) Chain

#### FOUNDATION

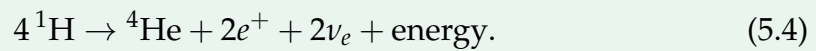
Representative net reaction:



#### Question Example 5.2

Calculate the energy released in the proton-proton chain reaction.

**Solution.** The net reaction for the proton-proton chain is:



The energy released can be calculated using the mass difference between the reactants and products:

$$\begin{aligned} \Delta m &= 4m_p - m_{^4\text{He}} - 2m_e \\ &\approx 4 \times 1.007276\ \text{u} - 4.002603\ \text{u} - 2 \times 0.0005486\ \text{u} \\ &\approx 0.0287\ \text{u}. \end{aligned}$$

Converting this mass difference to energy using  $E = \Delta mc^2$  and knowing that  $1\ \text{u} \approx 931.5\ \text{MeV}/c^2$ , we get:

$$\begin{aligned} E &= 0.0287\ \text{u} \times 931.5\ \text{MeV}/\text{u} \\ &\approx 26.7\ \text{MeV}. \end{aligned}$$

Thus, the energy released in the proton-proton chain reaction is approximately 26.7 MeV and carried away by neutrinos.

## 5.3 CNO Cycle

## 5.4 Thermonuclear Reaction Rates

### 5.4.1 Coulomb Barrier and Tunneling

**FOUNDATION**

The **Gamov energy** is the minimum kinetic energy that two nuclei must have in order to overcome the Coulomb barrier and undergo fusion. It is given by the formula:

$$E_G = (\pi\alpha Z_1 Z_2)^2 2m_r c^2 \quad (5.5)$$

where  $m_r$  is the reduced mass of the two nuclei,  $\alpha$  is the fine-structure constant, and  $Z_1$  and  $Z_2$  are the atomic numbers of the two nuclei.

**Question Example 5.3: The fusion of two protons**

Calculate the Gamov energy  $E_G$  for the fusion of two protons.

**Solution.** For two protons, we have  $Z_1 = Z_2 = 1$  and the reduced mass  $m_r$  is approximately equal to the mass of a proton,  $m_p$ . The fine-structure constant  $\alpha$  is approximately  $1/137$ . Plugging these values into the formula for the Gamov energy, we get:

$$\begin{aligned} E_G &= (\pi\alpha Z_1 Z_2)^2 2m_r c^2 \\ &= \left(\pi \times \frac{1}{137} \times 1 \times 1\right)^2 \times 2m_p c^2 \\ &= \left(\frac{\pi}{137}\right)^2 \times 2m_p c^2 \\ &\approx 493 \text{ keV}. \end{aligned}$$

Thus, the Gamov energy for the fusion of two protons is approximately 493 keV.

**Question Example 5.4: The fusion of carbon and oxygen**

Calculate the Gamov energy  $E_G$  for the fusion of carbon-12 ( $^{12}\text{C}$ ) and oxygen-16 ( $^{16}\text{O}$ ).

**Solution.** For carbon-12, we have  $Z_1 = 6$  and for oxygen-16, we have  $Z_2 = 8$ . The reduced mass  $m_r$  can be calculated using the formula:

$$m_r = \frac{m_1 m_2}{m_1 + m_2}, \quad (5.6)$$

where  $m_1$  and  $m_2$  are the masses of the carbon and oxygen nuclei, respectively. The mass of carbon-12 is approximately  $12m_p$  and the mass of oxygen-16 is

approximately  $16m_p$ . Thus, we have:

$$\begin{aligned} m_r &= \frac{(12m_p)(16m_p)}{12m_p + 16m_p} \\ &= \frac{192m_p^2}{28m_p} \\ &= \frac{192}{28}m_p \\ &\approx 6.86m_p. \end{aligned}$$

Now we can calculate the Gamov energy using the formula:

$$\begin{aligned} E_G &= (\pi\alpha Z_1 Z_2)^2 2m_r c^2 \\ &= \left(\pi \times \frac{1}{137} \times 6 \times 8\right)^2 \times 2 \times 6.86m_p c^2 \\ &= \left(\frac{48\pi}{137}\right)^2 \times 2 \times 6.86m_p c^2 \\ &\approx 8.7 \text{ MeV}. \end{aligned}$$

Thus, the Gamov energy for the fusion of carbon-12 and oxygen-16 is approximately 8.7 MeV.

#### FOUNDATION

The electrostatic interaction (Coulomb barrier) between two positively charged nuclei is given by:

$$V_C(r) = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 r}. \quad (5.7)$$

Classically, nuclei with  $E < V_C$  cannot fuse as they do not have enough kinetic energy to overcome the Coulomb barrier.

Quantum tunneling gives non-zero penetration probability  $P$  for  $E < V_C$ . The probability of tunneling through the Coulomb barrier can be approximated by the formula:

$$P \propto \exp \left[ -\sqrt{\frac{E_G}{E}} \right], \quad (5.8)$$

where  $E_G$  is the Gamow energy.

## 5.5 Thermonuclear Reaction Rates

## THEORY

$$\langle \sigma v \rangle = \left( \frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(k_B T)^{3/2}} \int_0^\infty S(E) \exp \left[ -\frac{E}{k_B T} - \sqrt{\frac{E_G}{E}} \right] dE. \quad (5.9)$$

The integrand peaks at the Gamow energy window, balancing high-energy tail abundance and tunneling probability.

## THEORY

$$R_{AB} \propto T^a \quad (5.10)$$

where  $a = \sqrt[3]{\frac{E_G}{4kT}}$  is a positive exponent that depends on the specific reaction and the temperature range.

## Question Example 5.5

Calculate the temperature dependence of the proton-proton (p-p) chain reaction rate.

**Solution.** The proton-proton chain reaction has a Gamow energy of approximately  $E_G \approx 493$  keV. The temperature dependence can be approximated by:

$$R_{pp} \propto T^{\sqrt[3]{\frac{E_G}{4kT}}}. \quad (5.11)$$

At typical stellar core temperatures of around  $T \approx 1.5 \times 10^7$  K, the exponent  $a$  can be calculated as:

$$a = \sqrt[3]{\frac{493 \text{ keV}}{4k(1.5 \times 10^7 \text{ K})}} \approx 4.5. \quad (5.12)$$

Thus, the reaction rate for the proton-proton chain scales approximately as  $R_{pp} \propto T^{4.5}$  at typical stellar core temperatures.

## 6.1 Stellar Structure Models

Main assumptions are made in simple stellar models:

- The star is in hydrostatic equilibrium.
- The star is in thermal equilibrium.
- The star is spherically symmetric.
- The star is non-rotating and has no magnetic fields.
- The star is composed of an ideal gas.

### THEORY

These assumptions allow us to derive the basic equations governing the structure of stars, such as the equations of hydrostatic equilibrium, mass conservation, energy generation, and energy transport. While these models are simplified, they provide valuable insights into the fundamental properties and behaviors of stars.

$$\text{Mass conservation : } \frac{dm(r)}{dr} = 4\pi r^2 \rho(r) \quad (6.1)$$

$$\text{Hydrostatic equilibrium : } \frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2} \quad (6.2)$$

$$\text{Thermal equilibrium : } L(r) = \int_0^r 4\pi r'^2 \rho(r') \epsilon(r') dr' \quad (6.3)$$

$$\text{Energy transport : } \frac{dT}{dr} = -\frac{3\kappa(r)\rho(r)L(r)}{16\pi acT^3 r^2} \quad (6.4)$$

### THEORY

By expressing the stellar structure equations in terms of mass  $m$  instead of radius  $r$ , we can derive a set of equations that are often more convenient for



numerical integration. The equations become:

$$\frac{dr}{dm} = \frac{1}{4\pi r^2 \rho} \quad (6.5)$$

$$\frac{dP}{dm} = -\frac{Gm}{4\pi r^4} \quad (6.6)$$

$$\frac{dL}{dm} = \epsilon \quad (6.7)$$

$$\frac{dT}{dm} = -\frac{3\kappa L}{16\pi acT^3 r^4} \quad (6.8)$$

### 6.1.1 Mass Range of Stars

The mass range of stars is typically defined as follows:

- **Low-mass stars:**  $M < 0.5M_{\odot}$  (e.g., red dwarfs)
- **Intermediate-mass stars:**  $0.5M_{\odot} < M < 8M_{\odot}$  (e.g., Sun-like stars)
- **High-mass stars:**  $M > 8M_{\odot}$  (e.g., O and B type stars)
- **Brown dwarfs:**  $M < 0.08M_{\odot}$  (substellar objects that do not sustain hydrogen fusion)

### 7.1 Hertzsprung-Russell (HR) Diagram

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### 7.2 Main Sequence Stars

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### 7.3 White Dwarfs

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#### THEORY

White dwarfs are supported by **electron degeneracy pressure**, which arises from the Pauli exclusion principle. The maximum mass of a white dwarf is given by the Chandrasekhar limit, approximately  $1.4M_{\odot}$ .

### 7.4 Supernovae

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## 8.1 Early Stages of Stellar Evolution

## 8.2 Late Stages of Stellar Evolution

### 8.2.1 Helium Flash

### 8.2.2 Giants and Supergiants

### 8.2.3 Type II Supernovae

## 8.3 Neutron Stars

### 8.3.1 Tolman-Oppenheimer-Volkoff (TOV) Equation

#### THEORY

The **Tolman-Oppenheimer-Volkoff (TOV) equation** describes the structure of a spherically symmetric body in hydrostatic equilibrium in general relativity. It is given by:

$$\frac{dP(r)}{dr} = - \frac{G \left( \rho(r) + \frac{P(r)}{c^2} \right) \left( m(r) + 4\pi r^3 \frac{P(r)}{c^2} \right)}{r^2 \left( 1 - \frac{2Gm(r)}{rc^2} \right)} \quad (8.1)$$

where  $P(r)$  is the pressure,  $\rho(r)$  is the energy density,  $m(r)$  is the mass enclosed within radius  $r$ ,  $G$  is the gravitational constant, and  $c$  is the speed of light. This equation is crucial for modeling neutron stars and other compact objects where relativistic effects are significant.

#### Question Example 8.1: Newtonian Limit of the TOV Equation

Show that in the Newtonian limit, the TOV equation reduces to the Lane-Emden equation for a polytropic fluid.

**Solution.** In the Newtonian limit, we assume that the pressure is much less than the energy density ( $P \ll \rho c^2$ ) and that the gravitational field is weak ( $\frac{2Gm(r)}{rc^2} \ll 1$ ). Under these assumptions, the TOV equation simplifies to:

$$\frac{dP(r)}{dr} \approx -\frac{G\rho(r)m(r)}{r^2} \quad (8.2)$$

This is the hydrostatic equilibrium equation in Newtonian gravity. For a polytropic fluid, we can express the pressure as a function of density:

$$P = K\rho^{1+\frac{1}{n}} \quad (8.3)$$

where  $K$  is a constant and  $n$  is the polytropic index. Substituting this into the hydrostatic equilibrium equation and introducing dimensionless variables leads to the Lane-Emden equation, which describes the structure of polytropic stars in Newtonian gravity.

### 8.3.2 Pulsars/X-ray Brusters

## 8.4 Black Holes

### 8.4.1 Schwarzschild Radius

#### THEORY

The **Schwarzschild radius** is the radius of the **event horizon** of a non-rotating black hole. It is given by:

$$r_s = \frac{2GM}{c^2} \quad (8.4)$$

where  $G$  is the gravitational constant,  $M$  is the mass of the black hole, and  $c$  is the speed of light. Any object that is compressed within its Schwarzschild radius will become a black hole.

#### Question Example 8.2: Schwarzschild Radius of a Collapsing Star

Suppose we have a star with a mass of  $10M_\odot$  (where  $M_\odot$  is the mass of the Sun). Calculate the Schwarzschild radius of the black hole that would form if this star were to collapse.

**Solution.** First, we need to convert the mass of the star into kilograms. The mass of the Sun is approximately  $1.989 \times 10^{30}$  kg, so:

$$M = 10M_\odot = 10 \times 1.989 \times 10^{30} \text{ kg} = 1.989 \times 10^{31} \text{ kg} \quad (8.5)$$

Using the formula for the Schwarzschild radius:

$$r_s = \frac{2GM}{c^2}, \quad (8.6)$$

we have

$$r_s = 29.49 \text{ km} \quad (8.7)$$

## 9.1 Chapter 1: Basic Concepts in Astrophysics

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### 9.1.1 Questions

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- 1.1 The star Gliese 710 is presently 62 lightyears from Earth, but Hipparcos data suggest that in about a million years it will pass within one lightyear of Earth. Its apparent magnitude is 9.7.
  - (a) What is the absolute magnitude of Gliese 710?
  - (b) What will be its apparent visual magnitude be in a million years as viewed from Earth?
- 1.2 Estimate the hydrodynamical free fall timescale for the Sun, a red giant of  $M = M_{\odot}$  and  $R = 100R_{\odot}$ , and a white dwarf of  $M = M_{\odot}$  and  $R = 0.01R_{\odot}$  and  $R = 100R_{\odot}$ , and a white dwarf of  $M = M_{\odot}$  and  $R = 0.02R_{\odot}$ .
- 1.3 Suppose that a star of mass  $M$  and radius  $R$  has a density distribution  $\rho(r) = \rho_c(1 - \frac{r}{R})$ , where  $\rho_c$  is the density at the center of the star.
  - (a) Find an expression for  $\rho_c$  in terms of  $M$  and  $R$ .
  - (b) Calculate the mass  $m(r)$  interior to radius  $r$ .
  - (c) Calculate the total gravitational binding energy of the star.
  - (d) Find an expression for the pressure  $P(r)$  as a function of  $r$ , assuming that the star is in hydrostatic equilibrium and that the pressure at the surface ( $r = R$ ) is zero.
  - (e) Assume that the material in the star is a **monatomic** ideal gas. Calculate the total internal energy of the star from  $P(r)$ , and show that the virial theorem is satisfied.

## 9.2 Chapter 2: Star Formation

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## 9.3 Chapter 3: Heat Transfers in Stars

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## 9.4 Chapter 4: Stellar Structure

## 9.5 Chapter 5: Nuclear Synthesis in Stars

### 9.5.1 Questions

4.1 In the simplest model of  $\alpha$  decay, the  $\alpha$  particle is pre-formed and trapped within the nucleus by a potential similar to a Coulomb barrier. The mean decay rate  $\lambda$  is the frequency  $\nu$  (with which the  $\alpha$  particle hits the barrier) times the probability of penetration of the barrier (quantum tunneling probability).

- (a) Write the approximate expression for the decay rate in terms of  $\nu$ ,  $E_G$ , and  $E$  (where  $E$  is the energy released by  $\alpha$  decay).
- (b) Find the Gamow energy of the  $\alpha$  decay of  $^{235}\text{U}$ .
- (c) Find the Gamow energy of the  $\alpha$  decay of  $^{239}\text{Pu}$ .
- (d) Assuming that the frequency  $\nu$  is approximately the same for both nuclei, find the ratio

$$\frac{\lambda(^{239}\text{Pu})}{\lambda(^{235}\text{U})}$$

using  $E(^{235}\text{U}) = 4.68\text{MeV}$  and  $E(^{239}\text{Pu}) = 5.24\text{MeV}$ . The lifetime of a radioactive species is inversely proportional to its decay rate. Given that the observed lifetime of  $^{235}\text{U}$  is  $\tau(^{235}\text{U}) = 7.1 \times 10^8 y$ , estimate the lifetime of  $^{239}\text{Pu}$ . Compare your result with the experimental value

$$\tau(^{239}\text{Pu}) = 2.41 \times 10^4 y$$

and comment on the accuracy of this simple model.

4.2 The **triple alpha process** is a set of nuclear reactions that occur in stars to produce carbon and other elements. Which of the following correctly describes the three steps of the triple alpha process, along with whether energy is absorbed or released during each step?

- (a)  $^4\text{He} + ^4\text{He} \rightarrow ^8\text{Be}$
- (b)  $^8\text{Be} + ^4\text{He} \rightarrow ^{12}\text{C}^*$
- (c)  $^{12}\text{C}^* \rightarrow ^{12}\text{C} + \gamma$

## 9.6 Chapter 7: Types of Stars

## 9.7 Chapter 8: Stellar Evolution

## A.1 Use of Artificial Intelligence in This Document

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This course guide was developed with assistance from artificial intelligence (AI) in the following capacities:

- **Template Design:** AI provided guidance on structuring the document layout, choosing appropriate color schemes.
- **LaTeX Code Generation:** AI assisted in writing and refactoring LaTeX code, including package configurations, box styling (tcolorbox), TikZ decorations, and custom command definitions.
- **LaTeX Structure Refactoring:** AI helped reorganize and optimize the document structure for maintainability, readability, and professional presentation.
- **Mathematical Formula Formatting:** AI provided assistance in formatting and presenting mathematical equations in clear, readable notation aligned with conventions.

**Author Verification:** All content, including scientific accuracy, theoretical derivations, example problems, and pedagogical choices, has been carefully reviewed and verified by the author. The AI assistance was limited to technical formatting and structural optimization, not to the generation or validation of course content itself.



## B.1 Useful Identities

### Trigonometric

$$\sin^2 \theta + \cos^2 \theta = 1, \quad \tan \theta = \sin \theta / \cos \theta, \quad \sin(A \pm B) = \sin A \cos B \pm \cos A \sin B.$$

### Hyperbolic

$$\cosh^2 x - \sinh^2 x = 1, \quad \sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y.$$

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

### Common Integrals

$$\begin{aligned} \int x^n dx &= \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), & \int e^{ax} dx &= \frac{1}{a} e^{ax} + C, \\ \int \frac{dx}{x} &= \ln |x| + C, & \int \sin x dx &= -\cos x + C. \end{aligned}$$

### Gamma-Zeta Integrals

$$\begin{aligned} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx &= \Gamma(s)\zeta(s), & \Re(s) &> 1, \\ \int_0^\infty \frac{x^{s-1}}{e^x + 1} dx &= (1 - 2^{1-s})\Gamma(s)\zeta(s), & \Re(s) &> 0. \end{aligned}$$

$$\begin{aligned} \int_0^\infty \frac{x^2}{e^x - 1} dx &= 2\zeta(3), & \int_0^\infty \frac{x^2}{e^x + 1} dx &= \frac{3}{2}\zeta(3), \\ \int_0^\infty \frac{x^3}{e^x - 1} dx &= \frac{\pi^4}{15} = 6\zeta(4), & \int_0^\infty \frac{x^3}{e^x + 1} dx &= \frac{7\pi^4}{120} = \frac{7}{8}6\zeta(4). \end{aligned}$$

## Taylor Expansions

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$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \dots$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \dots$$

$$\sqrt{1+x} = \sum_{n=0}^{\infty} \binom{1/2}{n} x^n = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \dots$$

## Common Approximations

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**Small-argument limits** ( $|x| \ll 1$ ):

$$\begin{array}{lll} \sin x \approx x, & \cos x \approx 1 - \frac{x^2}{2}, & \tan x \approx x, \\ \sinh x \approx x, & \cosh x \approx 1 + \frac{x^2}{2}, & \tanh x \approx x. \end{array}$$

**Large positive argument** ( $x \gg 1$ ):

$$\sinh x \approx \frac{1}{2}e^x, \quad \cosh x \approx \frac{1}{2}e^x, \quad \tanh x \approx 1.$$

**Large negative argument** ( $x \ll -1$ ):

$$\sinh x \approx -\frac{1}{2}e^{-x}, \quad \cosh x \approx \frac{1}{2}e^{-x}, \quad \tanh x \approx -1.$$


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## B.2 Relativistic Limits

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The energy-momentum relation is  $E^2 = p^2 c^2 + m^2 c^4$ . So in a relativistic limit, we can have:

$$\{E, m, \rho\} \ll \{p, T\}$$

which any one of the values in left-hand set is much smaller than any one of the values in the right-hand set.

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## C.1 Physical Constants

| Quantity                      | Symbol          | Value (SI)                                                         |
|-------------------------------|-----------------|--------------------------------------------------------------------|
| Speed of light                | $c$             | $2.998 \times 10^8 \text{ m s}^{-1}$                               |
| Gravitational constant        | $G$             | $6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ |
| Hubble constant today         | $H_0$           | $67.8 \pm 0.77 \text{ km s}^{-1} \text{ Mpc}^{-1}$                 |
| Cosmological constant         | $\Lambda$       | $1.11 \times 10^{-52} \text{ m}^{-2}$                              |
| Cosmological constant         | $\Lambda$       | $2.036 \times 10^{-35} \text{ s}^{-2}$                             |
| Planck constant               | $\hbar$         | $1.055 \times 10^{-34} \text{ J s}$                                |
| Boltzmann constant            | $k_B$           | $1.381 \times 10^{-23} \text{ J K}^{-1}$                           |
| Fermi constant                | $G_F$           | $1.166 \times 10^{-5} \text{ GeV}^{-2}$                            |
| Fine-structure constant today | $\alpha$        | $1/137.036$                                                        |
| Planck mass                   | $M_{\text{Pl}}$ | $2.176 \times 10^{-8} \text{ kg}$                                  |

## C.2 Conversion Factors

### Astronomical Units

| Conversion                                | Value                             |
|-------------------------------------------|-----------------------------------|
| 1 astronomical unit (AU)                  | $1.496 \times 10^{11} \text{ m}$  |
| 1 light-year (ly)                         | $9.461 \times 10^{15} \text{ m}$  |
| 1 parsec (pc)                             | $3.086 \times 10^{16} \text{ m}$  |
| 1 parsec (pc)                             | $3.262 \text{ ly}$                |
| 1 year (yr)                               | $3.156 \times 10^7 \text{ s}$     |
| 1 Earth mass ( $M_{\oplus}$ )             | $5.972 \times 10^{24} \text{ kg}$ |
| 1 Solar mass ( $M_{\odot}$ )              | $1.989 \times 10^{30} \text{ kg}$ |
| Milky Way' mass ( $M_{\text{MW}}$ )       | $1.5 \times 10^{12} M_{\odot}$    |
| 1 Earth radius ( $R_{\oplus}$ )           | $6.371 \times 10^6 \text{ m}$     |
| 1 solar radius ( $R_{\odot}$ )            | $6.957 \times 10^8 \text{ m}$     |
| Milky Way' radius                         | $15 \text{ kpc}$                  |
| 1 solar luminosity ( $L_{\odot}$ )        | $3.828 \times 10^{26} \text{ W}$  |
| Milky Way' luminosity ( $L_{\text{MW}}$ ) | $3.5 \times 10^{10} L_{\odot}$    |

Energy Units

| Conversion                | Value                              |
|---------------------------|------------------------------------|
| 1 eV                      | $1.602 \times 10^{-19} \text{ J}$  |
| 1 electron mass ( $m_e$ ) | $0.511 \text{ MeV}/c^2$            |
| 1 electron mass ( $m_e$ ) | $9.109 \times 10^{-31} \text{ kg}$ |
| 1 proton mass ( $m_p$ )   | $938.3 \text{ MeV}/c^2$            |
| 1 proton mass ( $m_p$ )   | $1.673 \times 10^{-27} \text{ kg}$ |
| 1 neutron mass ( $m_n$ )  | $939.6 \text{ MeV}/c^2$            |
| 1 neutron mass ( $m_n$ )  | $1.675 \times 10^{-27} \text{ kg}$ |

C.3 SI Unit Prefixes

| Prefix     | Name            | Factor     |
|------------|-----------------|------------|
| $10^{-12}$ | pico (p)        | $10^{-12}$ |
| $10^{-9}$  | nano (n)        | $10^{-9}$  |
| $10^{-6}$  | micro ( $\mu$ ) | $10^{-6}$  |
| $10^{-3}$  | milli (m)       | $10^{-3}$  |
| $10^3$     | kilo (k)        | $10^3$     |
| $10^6$     | mega (M)        | $10^6$     |
| $10^9$     | giga (G)        | $10^9$     |
| $10^{12}$  | tera (T)        | $10^{12}$  |

## C.4 Nuclei Binding Energies & Atomic Masses

| Nucleus          | Atomic Mass (u) | B(MeV)     | B/A (MeV/u) |
|------------------|-----------------|------------|-------------|
| $e^-$            | 0.00054858      | —          | —           |
| $p$              | 1.0072765       | —          | —           |
| $n$              | 1.0086649       | —          | —           |
| $^1\text{H}$     | 1.0078250       | 0.000000   | 0.000000    |
| $^2\text{H}$     | 2.0141018       | 2.224515   | 1.112258    |
| $^3\text{H}$     | 3.0160493       | 8.481773   | 2.827258    |
| $^3\text{He}$    | 3.0160293       | 7.718036   | 2.572679    |
| $^4\text{He}$    | 4.0026032       | 28.295803  | 7.073951    |
| $^6\text{Li}$    | 6.0151223       | 31.994603  | 5.332434    |
| $^7\text{Li}$    | 7.0160040       | 39.244654  | 5.606379    |
| $^8\text{Be}$    | 8.0053051       | 56.499667  | 7.062458    |
| $^9\text{Be}$    | 9.0121821       | 58.165096  | 6.462788    |
| $^{12}\text{C}$  | 12.0000000      | 92.162051  | 7.680171    |
| $^{13}\text{C}$  | 13.0033550      | 97.108223  | 7.469863    |
| $^{14}\text{N}$  | 14.0030740      | 104.658962 | 7.475640    |
| $^{15}\text{N}$  | 15.0001090      | 115.492214 | 7.699481    |
| $^{16}\text{O}$  | 15.9949150      | 127.619412 | 7.976213    |
| $^{17}\text{O}$  | 16.9991320      | 131.762631 | 7.750743    |
| $^{18}\text{O}$  | 17.9991610      | 139.806972 | 7.767054    |
| $^{20}\text{Ne}$ | 19.9924400      | 160.645559 | 8.032278    |
| $^{24}\text{Mg}$ | 23.9850420      | 198.257479 | 8.260728    |
| $^{28}\text{Si}$ | 27.9769270      | 236.537285 | 8.447760    |
| $^{32}\text{S}$  | 31.9720710      | 271.781333 | 8.493167    |
| $^{56}\text{Fe}$ | 55.9349420      | 492.255833 | 8.790283    |

Where **B** is the total binding energy of the nucleus, and **B/A** is the average binding energy per nucleon.

Note:  $u = 1.661 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV}/c^2$ .

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## REFERENCES

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